Minsung Jang

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AI Systems & AI Accelerators · Full-Stack AI Infrastructure for Efficient and Scalable AI

RESEARCH PROFILE

I build the computing systems that make large-scale AI practical, efficient, and trustworthy. My research spans **full-stack AI infrastructure systems**—GPU cluster management, LLM inference serving, AI datacenter networking, cloud systems, operating systems and virtualization, and **accelerator-aware runtime optimization** (processing-in-memory and low-precision kernels). I lead systems research that turns these ideas into real platforms and top-tier publications, and I frequently serve as a senior or research-leading co-author on recent Samsung SDS–led AI-systems work, while also contributing as an industry collaborator to joint academic research. My background bridges academic systems research and large-scale industrial research leadership (AT&T Labs Research; Samsung SDS Cloud Research Team), and I direct collaborations with university partners on heterogeneous-GPU training, intercloud orchestration, and confidential computing for secure AI execution.

RESEARCH INTERESTS

AI Infrastructure Systems · **Systems for Machine Learning** · **LLM Inference Serving** · **GPU Cluster Systems** · **AI Datacenter Networking** (RDMA/RoCE) · **Cloud Systems** · **Accelerator-aware Runtime Optimization** (PIM, quantization) · **Secure & Confidential AI Execution**

EDUCATION

Ph.D., Computer Science
Georgia Institute of Technology, Atlanta, GA, USA — August 2008 – August 2015
Dissertation: *Virtual Platforms: System Support to Enrich the Functionality of End Client Devices*
Advisor: Prof. Karsten Schwan

M.S., Mechanical Engineering
Yonsei University, Seoul, Korea — February 2000

B.S., Mechanical Design and Production Engineering
Yonsei University, Seoul, Korea — February 1998

SELECTED RESEARCH OUTPUTS

Names underlined for the candidate. Full list with all co-authors follows. Selected for fit to AI Systems / AI Accelerators.

- **PAISE** — PIM-Accelerated Inference Scheduling Engine for Transformer-based LLM.
  H. Lee, D. Baek, J. Son, J. Choi, K. Moon, <u>M. Jang</u>. **IEEE HPCA 2025**.
  GPU–PIM scheduler that offloads memory-bound attention to HBM-PIM, reducing LLM inference time by up to 48.3% versus GPU-only execution.
- **JABAS** — Joint Adaptive Batching and Automatic Scaling for DNN Training on Heterogeneous GPUs.
  G. Yun, J. Kang, H. Jeong, S. Eom, <u>M. Jang</u>, Y.-r. Choi. **EuroSys 2025**. (collab. with UNIST)
  Jointly tunes global batch size and GPU allocation, cutting training time by 33.3% and cost by 54.2% on heterogeneous GPU clusters with no accuracy loss.
- **DSDE** — Dynamic Speculative Decoding with KLD Stability for Real-World Serving.
  M. Yang, J.-Y. Choi, K. Moon, <u>M. Jang</u>, E. Jeon.
  IEEE BigData 2025, 11th Special Session on Intelligent Data Mining, pp. 3115–3124.
  Training-free, KLD-stability-driven adaptive speculation length that stays robust in low-acceptance-rate serving regimes.

- **CORN** — Cloud-optimized RDMA Networking.
  J.-H. Cha, S. Kang, Y. Kang, H. Seo, J. Lee, J. Kim, M. Jang. **IEEE IPCCC 2023**.
  End-host RoCEv2 congestion control and adaptive multipathing for GPU interconnect that removes dependence on switch-level PFC.
- **Omni-MP** — Practical PIM Matrix Mapping for Accelerating LLM Inference on Real PIM Hardware.
  D. Baek, J. Son, J. Choi, K. Bin, S. Choi, K. Moon, M. Jang, H. Lee. **MCCSys @ ACM ICS 2026** (to appear).
  Generalized PIM-GEMV mapping that reshapes arbitrary LLM weight matrices and KV-cache to fit fixed PIM tiles—covering attention, not just FFN—for 40.2% faster end-to-end inference on average versus GPU-only on real HBM-PIM.
- **FineServe** — Precision-Aware KV Slab and Two-Level Scheduling for Heterogeneous Precision LLM Serving.
  K. Bin, S. Choi, J. Son, J. Choi, D. Bae, D. Baek, K. Moon, M. Jang, H. Lee. **Preprint, under review** (arXiv:2509.06261).
  Precision-aware KV-cache management and two-level scheduling for mixed-precision LLMs; up to 2.2× higher SLO attainment and 1.8× higher throughput.

PUBLICATIONS

Candidate name underlined. Senior/corresponding-author role on much of the recent AI-systems work.

Refereed Conference & Workshop Publications

- H. Lee, D. Baek, J. Son, J. Choi, K. Moon, M. Jang. *PAISE: PIM-Accelerated Inference Scheduling Engine for Transformer-based LLM.* Proceedings of the 2025 IEEE International Symposium on High-Performance Computer Architecture (HPCA), 2025.
- G. Yun, J. Kang, H. Jeong, S. Eom, M. Jang, Y.-r. Choi. *JABAS: Joint Adaptive Batching and Automatic Scaling for DNN Training on Heterogeneous GPUs.* Proceedings of the Twentieth European Conference on Computer Systems (EuroSys), 2025.
- M. Yang, J.-Y. Choi, K. Moon, M. Jang, E. Jeon. *DSDE: Dynamic Speculative Decoding with KLD Stability for Real-World Serving.* Proceedings of the 2025 IEEE International Conference on Big Data (IEEE BigData), 11th Special Session on Intelligent Data Mining, pp. 3115–3124, 2025.
- D. Baek, J. Son, J. Choi, K. Bin, S. Choi, K. Moon, M. Jang, H. Lee. *Omni-MP: Practical PIM Matrix Mapping for Accelerating LLM Inference on Real PIM Hardware* (to appear). 7th Workshop on Memory-Centric Computing Systems (MCCSys), in conjunction with the ACM International Conference on Supercomputing (ICS), 2026.
- H. Lee, L. Vu, M. Jang. *Economics of Spot Instance Service: A Two-Stage Dynamic Game Approach.* IEEE International Conference on Cloud Computing (IEEE CLOUD), 2023.
- J.-H. Cha, S. Kang, Y. Kang, H. Seo, J. Lee, J. Kim, M. Jang. *CORN: Cloud-optimized RDMA Networking.* IEEE International Performance, Computing, and Communications Conference (IPCCC), 2023.
- R. M. Krishnan, W.-H. Kim, X. Fu, S. K. Monga, H. W. Lee, M. Jang, A. Mathew, C. Min. *TIPS: Making Volatile Index Structures Persistent with DRAM-NVMM Tiering.* USENIX Annual Technical Conference (USENIX ATC), 2021.
- E. F. Boza, C. L. Abad, S. P. Narayanan, B. Balasubramanian, M. Jang. *A Case for Performance-Aware Deployment of Containers.* 5th International Workshop on Container Technologies and Container Clouds (WoC), 2019.
- H. Gupta, A. Sharma, A. Zelezniak, M. Jang, U. Ramachandran. *A Black-Box Approach for Estimating Utilization of Polled IO Network Functions.* 11th USENIX Workshop on Hot Topics in Cloud Computing (HotCloud), 2019.
- M. Jang, H. Lee, K. Bhardwaj, K. Schwan. *SOUL: An Edge-Cloud System for Mobile Applications in a Sensor-Rich World.* IEEE/ACM Symposium on Edge Computing (SEC), 2016.
- M. Jang, H. Lee, K. Bhardwaj, K. Schwan. *vSensor: Toward Sensor-rich Mobile Applications.* Sensors to Cloud Architectures Workshop (SCAW), in conjunction with HPCA-21, 2015.

- M. Jang, K. Schwan, K. Bhardwaj, A. Gavrilovska, A. Avasthi. *Personal Clouds: Sharing and Integrating Networked Resources to Enhance End User Experiences.* IEEE International Conference on Computer Communications (IEEE INFOCOM), 2014. (Acceptance rate: 19.4%)
- M. Jang, K. Schwan. *Clouds4Users: From Isolated Devices to Rich Device Platforms* (poster). 3rd Workshop on SoCs, Heterogeneous Architectures and Workloads (SHAW-3), in conjunction with HPCA-18, 2012.
- M. Jang, K. Schwan. *STRATUS: Assembling Virtual Platforms from Device Clouds.* IEEE International Conference on Cloud Computing (IEEE CLOUD), 2011. (Acceptance rate: 18%)
- S. Heo, M. Jang, S. Suh. *Software Mobility for Personalized Computing Environments on Removable Storage.* IEEE Consumer Communications and Networking Conference (IEEE CCNC), 2008.
- J. Hwang, J. Bae, A. Kirnasov, M. Jang, H. Kim. *A Reliable and Portable Multimedia File System.* Linux Symposium, 2006.

Preprints / Under Review

- K. Bin, S. Choi, J. Son, J. Choi, D. Bae, D. Baek, K. Moon, M. Jang, H. Lee. *FineServe: Precision-Aware KV Slab and Two-Level Scheduling for Heterogeneous Precision LLM Serving.* Preprint (under review), arXiv:2509.06261, 2025.
- D. Baek, J. Choi, J. Son, K. Bin, S. Choi, K. Moon, M. Jang, H. Lee. *FireQ: Fast INT4-FP8 Kernel and RoPE-aware Quantization for LLM Inference Acceleration.* Preprint (under review), arXiv:2505.20839, 2025.
- S. Lee, B. Woo, H. Kang, H. Choe, H. Lee, K. Moon, M. Jang, B. Kang. *Lightweight and Efficient Nested Enclaves for AMD SEV-SNP Confidential VMs.* Under review
- S. Choi, J. Goo, E. Jeon, M. Yang, M. Jang. *ELIS: Efficient LLM Iterative Scheduling System with Response Length Predictor.* Preprint, arXiv:2505.09142, 2025.
- S. Kim, Y. Kim, K. Moon, M. Jang. *LaDiMo: Layer-wise Distillation Inspired MoEfier.* Preprint, arXiv:2408.04278, 2024.

RESEARCH & TECHNICAL LEADERSHIP

Samsung SDS August 2021 – Present

Executive Advisor (VP-level), Seoul, Korea January 2026 – Present

Vice President & Head, Cloud Research Team, Seoul, Korea August 2021 – December 2025

- **Directed a full-stack AI-infrastructure research organization** — the Cloud Research Team, structured around three labs (computing systems, networks, and high-performance computing) and 65+ researchers and engineers across Seoul, Mountain View, and Seattle. Set the research agenda spanning GPU cluster systems, LLM inference serving, AI datacenter networking, and accelerator-aware runtimes.
- **Built a publishing systems-research program** whose results appear at **HPCA** (PAISE), **IEEE BigData** (DSDE), **MCCSys@ICS** (Omni-MP), and as preprints under review (FireQ, FineServe); served as senior/corresponding author on much of this work.
- **Accelerator-aware inference research:** led a processing-in-memory (PIM) program on real HBM-PIM hardware (AMD Instinct MI100), producing GPU–PIM inference scheduling (PAISE) and practical PIM matrix mapping (Omni-MP), alongside low-precision INT4–FP8 kernels (FireQ) and precision-aware serving (FineServe).
- **GPU-as-a-Service & LLM serving:** architected Samsung's Kubernetes-based GPU-as-a-Service on Samsung Cloud Platform (SCP)—GPU-aware scheduling, multi-tenant isolation/quotas, inference endpoints, observability, and cost controls—improving GPU utilization and lowering inference cost for enterprise AI workloads.
- **AI training/inference platform:** led design and deployment of **x.Cloud**, a Kubernetes-based distributed training and inference platform (presented at NVIDIA GTC 2021, "High-performance/High-efficiency AI Model Training Cluster Based on Kubernetes").
- **AI datacenter networking:** directed **CORN** (Cloud-optimized RDMA Networking), an end-host RoCEv2 congestion-control and adaptive multipath design for GPU interconnect (IPCCC 2023), subsequently implemented as a hardware-NIC prototype in collaboration with **AMD** and demonstrated

at **SC25**; Samsung representative to the **Ultra Ethernet Consortium (UEC)** on next-generation AI-datacenter networks.
- **Cloud, OS & storage systems:** technology lead for Samsung Cloud Platform (OpenStack-based open-architecture cloud) and chief architect of its vendor-agnostic storage control plane (performance/capacity-aware volume placement, archival storage).
- **University research collaborations:** represented and led Samsung SDS's participation as a founding sponsor of the UC Berkeley Sky Computing Lab, with joint research in Prof. Ion Stoica's team on AI datacenter network, LLM serving, and intercloud orchestration; UNIST (Prof. Young-ri Choi) on heterogeneous-GPU training (EuroSys 2025); KAIST Cyber Security Systems Research Lab (Prof. Brent Byunghoon Kang) on confidential computing for secure AI execution.

Peraton Labs (formerly Perspecta Labs) September 2020 – July 2021
Senior Research Scientist, Basking Ridge, NJ, USA
- Applied research on networking and distributed systems under performance, resilience, and operational constraints.

AT&T Labs Research August 2015 – July 2020
Principal-Inventive Scientist, Bedminster, NJ, USA
- **GPU-accelerated networking systems:** architected a GPU-based NFV data plane to accelerate stateful, compute-intensive network functions to NVIDIA GPUs (presented at NVIDIA GTC 2018, "Practical GPU-Based Network Packet Processing"); foundation for granted patents.
- **Accelerator abstraction for high-performance software-defined network:** led R&D on cloud-native network services (5G and Internet) on the commodity server with hardware acceleration (e.g., smartNICs, FPGAs, and ASICs); contributed to the **O-RAN Alliance Working Group 6** on accelerator abstraction and O-Cloud architecture.
- **Persistent-memory systems:** co-designed and evaluated a persistent-memory key-value store for telco workloads (Intel Optane), contributing to systems research published at USENIX ATC.
- **Cloud-native infrastructure:** designed and implemented the AT&T Network Cloud for carrier-grade, nationwide (United States) 5G and Internet services.

Samsung Electronics & Samsung Advanced Institute of Technology August 2003 – August 2008
Research Staff Member, Kiheung, Korea
- Built **XenARM**, hypervisor technology to virtualize ARM-based platforms for embedded devices; systems-software and virtualization research for next-generation mobile/consumer platforms.

GRADUATE RESEARCH INTERNSHIPS

- **Intel Labs**, Integrated Platform Research, Hillsboro, OR Summer 2012
- **Nokia Research Center**, North America Research, Palo Alto, CA Summer 2011
- **IBM Almaden Research Center**, Storage Systems Group, San Jose, CA Summer 2010

RESEARCH COLLABORATIONS & PROFESSIONAL SERVICE

- **UC Berkeley Sky Computing Lab** (Prof. Ion Stoica) — represented and led Samsung SDS's participation as a founding sponsor; joint research with Prof. Stoica's team on AI infrastructure systems and intercloud orchestration.
- **UNIST** (Prof. Young-ri Choi) — adaptive batching and autoscaling for DNN training on heterogeneous GPUs (JABAS, EuroSys 2025).
- **KAIST Cyber Security Systems Research Lab** (Prof. Brent Byunghoon Kang) — confidential computing for secure AI execution: lightweight nested enclaves on AMD SEV-SNP (under review).
- **AMD** — **CORN** implemented as a hardware-NIC prototype and demonstrated at **SC25 (Supercomputing 2025)**
- **Intel** — jointly improved the AI system software stack for the **Gaudi 3 AI accelerator** platform and presented it at **Intel AI Summit 2025**.

- **Korean AI Accelerators (NPUs) Performance-Evaluation Initiative (K-Perf)** — represented Samsung SDS as the demand-side cloud-service-provider participant in this Ministry of Science and ICT / NIPA-led public–private initiative establishing industry-wide performance-evaluation metrics for Korean NPUs; contributed benchmark conditions and metrics for real-world LLM serving (concurrency, latency, throughput, power efficiency).
- **Ultra Ethernet Consortium (UEC)** — Samsung representative, next-generation networking for AI datacenters November 2023 – Present
- **O-RAN Alliance**, Working Group 6 (O-Cloud / accelerator abstraction) — AT&T representative and contributor 2019 – 2020

INVITED TALKS (selected)

- **SC25 (Supercomputing 2025)**, Technical Contributor, Co-presentation with AMD — CORN (Cloud-optimized RDMA Networking) implemented as a hardware-NIC prototype, November 2025.
- **Intel AI Summit Seoul 2025**, Invited Speaker — "Performance Analysis of Intel Gaudi 3 for LLM Inference," Seoul, July 2025.
- **Seoul National University**, Invited Lecture — "Trends in AI Infrastructure Technologies," Seoul, September 2024.
- **Korea Computer Congress (KCC) 2024**, Invited Speaker — "Cloud Technologies and Vision Behind Generative AI," Jeju, June 2024.
- **UC Berkeley Sky Computing Retreats 2024**, Collaboration Partner Speaker — "SkyAirFlow: Sky-powered Intercloud Workload Orchestration" (May 2024) and "Samsung Cloud Platform's Integration with Sky" (Jan. 2024).
- **High Speed Network (HSN) 2024**, Invited Speaker — "Emerging Technologies in Cloud Computing 2024 and Beyond," Jeju, January 2024.
- **KIISE Computer System Society Winter Workshop 2023**, Invited Speaker — "x.Cloud: Samsung SDS GPU-based AI Development Platform," Pyeongchang, February 2023.
- **NVIDIA GTC 2021**, Co-speaker — "High-performance/High-efficiency AI Model Training Cluster Based on Kubernetes (x.Cloud)," November 2021.
- **NVIDIA GTC 2018**, Technical Contributor — "Practical GPU-Based Network Packet Processing: Building the Next Generation Internet Router," San Jose, CA.
- **IEEE/ACM Symposium on Edge Computing** (2016), **IEEE INFOCOM** (2014), **IEEE CLOUD** (2011); research talks at **IBM Almaden** (2014) and **AT&T Labs Research** (2015).

SELECTED PATENTS

- **Persistent kernel for GPU direct-memory-access network packet processing** — co-inventor; persistent GPU kernels + DMA for high-throughput NFV packet processing. (US 10,795,840 and related family)
- **Direct memory access for GPU packet processing** — co-inventor; efficient NIC–GPU DMA to reduce data movement for GPU-based network functions. (US 10,332,235 and related family)
- **Scaling network capability using baseband unit pooling in 5G and beyond** — co-inventor; BBU pooling and dynamic resource allocation for cloud-native vRAN. (US 11,582,642 and related family)
- **Operating a mobile virtual environment upon connection to a host computer** — co-inventor; running personal virtual environments from removable storage via virtualization. (US 10,007,541 and related family)
- **Managing file-system data using a database management system** — co-inventor; DBMS-backed file-system metadata for reliability and scalability. (US 9,384,201 and related family)

TEACHING & MENTORING

- **Guest Lecturer**, CS7210 Distributed Systems, Georgia Tech Spring 2013
- **Graduate Teaching Assistant**, CS6210 Advanced Operating Systems, Georgia Tech Spring 2012
- **University lectures** on AI infrastructure and cloud/systems topics (Seoul National University, 2024; Korea Computer Congress, 2024).

- **Doctoral mentoring:** Hyunjong Lee (Ph.D.). **M.S. mentoring:** Vaibhav Bedia, Rajaram Shetty, Aneesh Mulye.
- **Research mentoring:** advised researchers and engineers in the Samsung SDS Cloud Research Team in producing first-authored top-tier systems papers; co-advised graduate-student collaborators at partner universities.

HONORS & AWARDS

- Intel Ph.D. Fellowship nominee, Georgia Institute of Technology, 2013–2014
- Intel Kudos Award, Intel CountryFair, Intel Corporation, 2012
- Excellent Research Project Award, Software Laboratories, Samsung Electronics, 2003

TECHNICAL EXPERTISE

AI/ML Systems: LLM inference serving (vLLM/KServe), GPU-aware scheduling, speculative decoding, precision-aware KV serving, quantization, PIM-accelerated inference, heterogeneous-GPU training, GPU-as-a-Service on Kubernetes

Accelerators & Performance: HBM-PIM, CUDA, GPUDirect RDMA, INT4/FP8 GEMM kernels, Nsight/NVTX profiling

Networking & Data Plane: RDMA/RoCEv2 congestion control, DPDK, XDP/eBPF, SDN, 5G vRAN, AI-datacenter networking

Cloud, OS & Storage: Kubernetes, OpenStack, virtualization/hypervisors, distributed systems, storage control planes, persistent-memory systems